多云底座常见问题处理手册

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1. 运行状态异常处理

1.1 多云底座控制台页面无法访问

1. 登录dce服务器，查看dce容器是否正在运行

```
[root@dce ~]# docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED      STATUS      PORTS
4c8da1f73254   dce:20220722-1600   "/usr/bin/supervisor…"   2 days ago   Up 2 days   0.0.0.0:9001->9001/tcp, :::9001->9001/tcp, 0.
96e54dfb9418   bogem/ftp           "/usr/sbin/run-vsftp…"   4 days ago   Up 4 days   0.0.0.0:47400-47470->47400-47470/tcp, :::4746
345b1bacd745   mysql:8.0.28        "docker-entrypoint.s…"   4 days ago   Up 4 days   0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 33
```

2. 如果无法查询到容器运行情况，检查docker服务是否正常运行

```
systemctl status docker
```

3. 如果dce容器正在运行，检查端口映射是否正常，默认映射本机8003端口

```
[root@dce ~]# netstat -an |grep 8003
tcp        0      0 0.0.0.0:8003            0.0.0.0:*               LISTEN
tcp6       0      0 :::8003                 :::*                    LISTEN
[root@dce ~]#
```

4. 端口监听正常，检查本机防火墙是否已关闭或开启访问策略

```
[root@dce ~]# systemctl  status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@dce ~]#
```

5. 检查容器运行日志，是否正常启动容器

```
[root@dce ~]# docker logs -f dce
2022-07-22 16:04:15,156 CRIT Supervisor is running as root.  Privileges were not dropped because no user is specified in the confi
2022-07-22 16:04:15,156 WARN For [program:api-server], redirect_stderr=true but stderr_logfile has also been set to a filename, th
2022-07-22 16:04:15,159 INFO supervisord started with pid 1
2022-07-22 16:04:16,162 INFO spawned: 'api-server' with pid 7
2022-07-22 16:04:16,164 INFO spawned: 'nginx' with pid 8
2022-07-22 16:04:17,287 INFO success: api-server entered RUNNING state, process has stayed up for > than 1 seconds (startsecs)
2022-07-22 16:04:17,287 INFO success: nginx entered RUNNING state, process has stayed up for > than 1 seconds (startsecs)
```

1.2 系统登录失败

1. 登录数据库服务器，查看mysql是否正常运行

```
[root@dce ~]# docker ps |grep mysql
345b1bacd745   mysql:8.0.28        "docker-entrypoint.s…"   4 days ago   Up 4 days      0.0.0.0:3306->3306/tcp, :::3306->3306/tcp,
[root@dce ~]#
```

2. 查看数据库端口是否正常监听

```
[root@dce ~]# netstat -an |grep 3306
tcp        0      0 0.0.0.0:3306            0.0.0.0:*               LISTEN
tcp        0      0 172.17.0.1:33980        172.17.0.2:3306         ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57806        ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57840        ESTABLISHED
tcp        0      0 172.17.0.1:33950        172.17.0.2:3306         ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57824        ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57802        ESTABLISHED
tcp        0      0 172.17.0.1:33968        172.17.0.2:3306         ESTABLISHED
tcp        0      0 172.17.0.1:33984        172.17.0.2:3306         ESTABLISHED
tcp        0      0 172.17.0.1:33946        172.17.0.2:3306         ESTABLISHED
tcp        0      0 172.17.0.1:34068        172.17.0.2:3306         ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57836        ESTABLISHED
tcp        0      0 10.101.9.44:3306        172.17.0.4:57924        ESTABLISHED
tcp6       0      0 :::3306                 :::*                    LISTEN
[root@dce ~]#
```

3. 进入数据库容器，使用dce 配置文件中的数据库账号密码登录数据库，查看是否可以查询表

```
[root@dce ~]# docker exec -it mysql bash
root@345b1bacd745:/# mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 67
Server version: 8.0.28 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use dscloud_1_0;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-------------------------------------+
| Tables_in_dscloud_1_0               |
+-------------------------------------+
| chart_stores                        |
| dsc_alarm_messages                  |
| dsc_alarm_rules                     |
| dsc_alert_notify_history            |
| dsc_alert_notify_receiver_group     |
| dsc_alert_notify_receiver_invitation |
| dsc_alert_notify_strategy           |
| dsc_alert_notify_strategy_rule      |
| dsc_app_package                     |
| dsc_audit                           |
| dsc_cluster_operation_log           |
| dsc_cluster_package                 |
| dsc_clusters                        |
| dsc_configs                         |
| dsc_labels                          |
| dsc_medium_repo                     |
| dsc_pulgin_templates                |
| dsc_storage                         |
| dsc_storage_cluster                 |
| dsc_storages                        |
| dsc_tenant_cluster                  |
| dsc_tenant_repository               |
| dsc_tenant_user                     |
| dsc_tenants                         |
| dsc_users                           |
+-------------------------------------+
25 rows in set (0.00 sec)

mysql>
```

4. 核对dce容器配置文件中的数据库连接地址，账号密码, 主要检查runmode对应的数据库配置和capacityDB配置信息， 密码可以使用加密工具进行密码解密

```
db:
  runmode: pro  # {dev | pro | public}
  gormlog: true   # 数据库日志
  db_max_idle: 10
  db_max_conn: 100
  dev:
    name: dscloud_1_0
    addr: 10.101.9.19:3306
    username: root
    password: jRuizRC7JdlXeBLA6DXaQOTZ6K6xSiqSEtkw6I3wiHs=
  pro:
    name: dscloud_1_0
    addr: 10.101.9.44:3306
    username: root
    password: q_4CKEgp88UpU13wLVhwjS8f663FqAD_-BRN_SbJ_ms=
  public:
    name: dscloud_1_0
    addr: 47.92.202.208:13306
    username: root
    password: jRuizRC7JdlXeBLA6DXaQOTZ6K6xSiqSEtkw6I3wiHs=
  capacityDB:
    name: information_schema
    addr: 10.101.9.44:3306
    username: root
    password: q_4CKEgp88UpU13wLVhwjS8f663FqAD_-BRN_SbJ_ms=
```

5. 如果数据库配置信息正常，可以查看dce日志进行问题排查

```
tailf /paas/dce/logs/api-server.log
```

1.3 镜像仓库无法查看

1. 查看镜像仓库服务器，harbor服务对应的容器是否正常运行

```
[root@harbor harbor]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED      STATUS                PORTS
dc9c0e7ed205   goharbor/harbor-jobservice:v2.4.2    "/harbor/entrypoint.…"   4 days ago   Up 4 days (healthy)
ef7b4254fa69   goharbor/nginx-photon:v2.4.2         "nginx -g 'daemon of…"   4 days ago   Up 4 days (healthy)   0.0.0.0:80->8080/t
95fa6ea16c9f   goharbor/harbor-core:v2.4.2          "/harbor/entrypoint.…"   4 days ago   Up 4 days (healthy)
428b28bf762f   goharbor/harbor-db:v2.4.2            "/docker-entrypoint.…"   4 days ago   Up 4 days (healthy)
00ee82afddc4   goharbor/harbor-registryctl:v2.4.2   "/home/harbor/start.…"   4 days ago   Up 4 days (healthy)
ec1ec4900e5d   goharbor/chartmuseum-photon:v2.4.2   "./docker-entrypoint…"   4 days ago   Up 4 days (healthy)
cc0c3a2dd57e   goharbor/redis-photon:v2.4.2         "redis-server /etc/r…"   4 days ago   Up 4 days (healthy)
bfa75331c21b   goharbor/registry-photon:v2.4.2      "/home/harbor/entryp…"   4 days ago   Up 4 days (healthy)
3c37ea3b6ec1   goharbor/harbor-portal:v2.4.2        "nginx -g 'daemon of…"   4 days ago   Up 4 days (healthy)
360d78b4325d   goharbor/harbor-log:v2.4.2           "/bin/sh -c /usr/loc…"   4 days ago   Up 4 days (healthy)   127.0.0.1:1514->16
[root@harbor harbor]#
```

2. 如果harbor服务未正常运行，执行启动命令进行服务重新启动

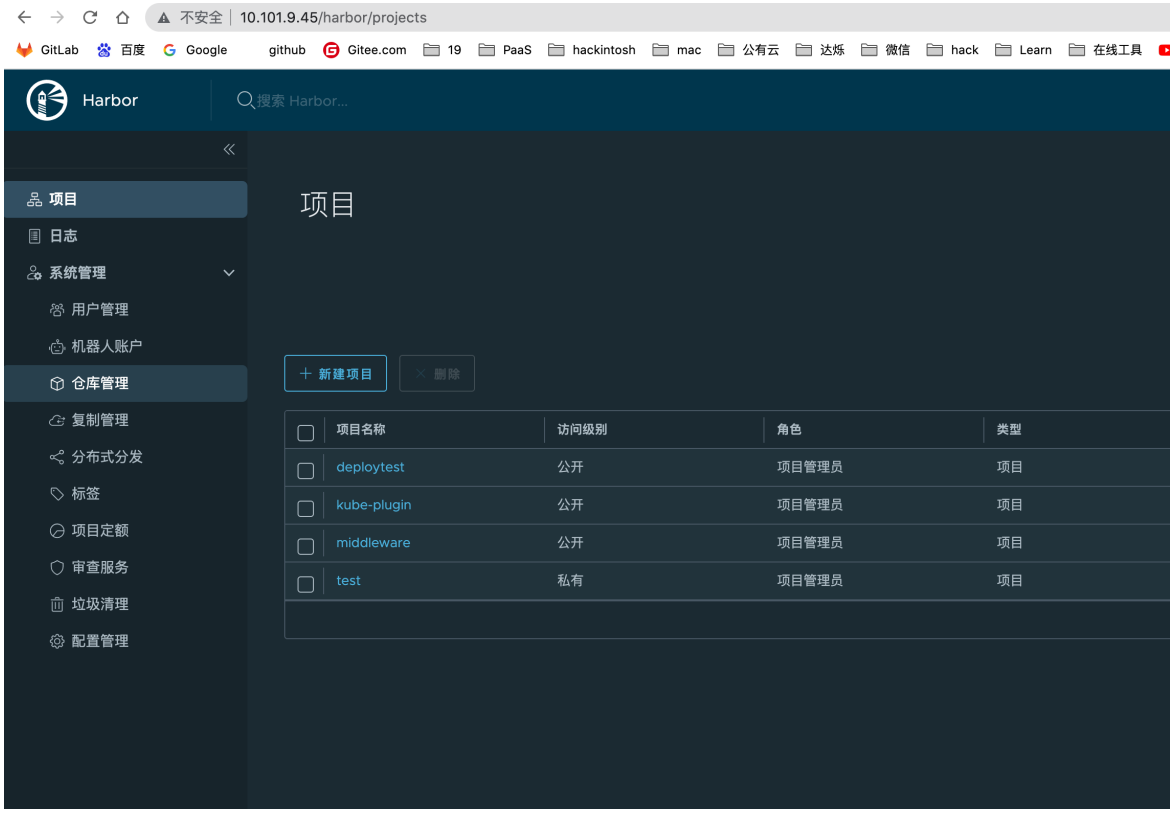
```
[root@harbor harbor]# cd /opt/harbor/
[root@harbor harbor]# docker-compose down -v
[root@harbor harbor]# docker-compose up -d
harbor-log is up-to-date
registry is up-to-date
harbor-portal is up-to-date
redis is up-to-date
harbor-db is up-to-date
registryctl is up-to-date
chartmuseum is up-to-date
harbor-core is up-to-date
harbor-jobservice is up-to-date
nginx is up-to-date
[root@harbor harbor]#
```

3. 检查服务器防火墙是否关闭

```
[root@harbor harbor]# systemctl  status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)

4月 24 20:06:23 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon...
4月 24 20:06:24 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
4月 24 20:06:24 localhost.localdomain firewalld[824]: WARNING: AllowZoneDrifting is enabled. This is consi...ow.
5月 04 22:44:10 node1 systemd[1]: Stopping firewalld - dynamic firewall daemon...
5月 04 22:44:12 node1 systemd[1]: Stopped firewalld - dynamic firewall daemon.
Hint: Some lines were ellipsized, use -l to show in full.
[root@harbor harbor]#
```

4. 访问harbor控制台界面，查看是否可以正常访问和登录



5. 检查多云底座 资源接入 - 介质仓库接入 配置的镜像仓库地址，账号密码是否正确

1.4 软件包上传下载失败

1. 登录文件服务器，查看文件服务器的容器是否正常运行

```
[root@dce dce]# docker ps |grep ftp
96e54dfb9418   bogem/ftp           "/usr/sbin/run-vsftp…"   4 days ago   Up 4 days       0.0.0.0:47400-47470->47400-47470/tcp, :::
[root@dce dce]#
```

2. 查看ftp服务监听的端口是否正常

```
[root@dce dce]# netstat -an|grep 10021
tcp        0      0 0.0.0.0:10021           0.0.0.0:*               LISTEN
tcp6       0      0 :::10021                ::::*                   LISTEN
[root@dce dce]#
```

3. 检查防火墙是否已关闭或者访问策略放行

```
[root@dce dce]# systemctl  status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@dce dce]#
```

4. 登录多云底座，检查 资源接入 介质仓库接入是否正确配置文件服务器的地址 端口 账号和密码

5. 文件服务器的账号密码在启动脚本中配置FTP_USER、FTP_PASS

```
[root@dce ~]# cat run-ftp
ftp_vip=10.101.9.44

docker run -v /root/ftp/upload:/home/vsftpd \
  -p 10020:20 -p 10021:21 -p 47400-47470:47400-47470 \
  -e FTP_USER=hengshi \
  -e FTP_PASS=2022.Com \
  -e PASV_ADDRESS=$ftp_vip \
  --name ftp \
  --restart=always \
  -d bogem/ftp
[root@dce ~]#
```

1.5 集群状态异常

1. 登录被管集群master节点，查看集群状态是否正常

```
[root@master ~]# kubectl get nodes
NAME     STATUS   ROLES                  AGE     VERSION
master   Ready    control-plane,master   2d17h   v1.20.9-dirty
node1    Ready    <none>                 2d17h   v1.20.9-dirty
node2    Ready    <none>                 2d17h   v1.20.9-dirty
[root@master ~]#
```

2. 如果集群节点状态异常或无法查看，请排查k8s集群故障
3. 检查集群接入填写的k8s apiserver地址端口，和token是否准确有效
4. 集群异常，详细错误信息可以查看dce日志

```
tailf /paas/dce/logs/api-server.log
```

5. 如果提示集群接口调用未认证获取无权限，使用命令创建token，并更新到集群接入表中dsc_clusters

```
kubectl edit clusterrolebinding cluster-admin
#添加以下内容
subjects:
- kind: ServiceAccount
  name: default
  namespace: default
- kind: ServiceAccount
  name: default
  namespace: kube-system

#获取token
kubectl describe secret default
```

2. 集群管理异常处理

2.1 集群插件安装失败

1. 登录集群master节点，执行命令查看是否存在helm-operator的Pod

```
[root@master ~]# kubectl get pod -A |grep helm-operator
flux          helm-operator-859f8448c7-mmmwf                   1/1     Running     0          2d17h
[root@master ~]#
```

2. 如果该Pod不存在，则需要上传安装文件helm-operator.yaml到master

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/4f9a4014-cdbd-408b-8302-5e6ca5b6535b/helm-operator.yaml

```
#修改文件中{{registry}}为 管理集群master1IP:5000
kubectl apply -f helm-operator.yaml
```

3. 如果helm-operator正常运行，安装插件失败，检查dce日志

```
tailf /paas/dce/logs/api-server.log
```

2.2 集群节点无性能监控数据

1. 登录多云底座，查看集群插件prometheus是否正常安装，如未安装，则先安装如下插件

```
metrics-server
node-exporter
prometheus
alertmanager
```

2. 查看安装完成后，登录master节点，查看插件对应的Pod是否都正常运行

```
[root@master ~]# kubectl  get pod -n kube-plugin
NAME                                          READY   STATUS    RESTARTS   AGE
alertmanager-5d8bf7bc5f-s7tmz                 1/1     Running   0          2d
```

```
application-controller-manager-546f5b97b-w4m5n   2/2     Running   0          2d
elastic-operator-0                               1/1     Running   1          2d
elasticsearch-es-nodes-0                         1/1     Running   0          27h
fluentd-lnw9b                                    1/1     Running   0          27h
fluentd-mjgdn                                    1/1     Running   0          27h
fluentd-qqpsj                                    1/1     Running   0          27h
ingress-controller-qgz92                         1/1     Running   0          2d
metrics-server-97b57bf7d-cm6b6                   1/1     Running   0          2d
node-exporter-glhlv                              1/1     Running   0          2d
node-exporter-lsttc                              1/1     Running   0          2d
node-exporter-r7lv7                              1/1     Running   0          2d
prometheus-6544694566-9zj2w                      2/2     Running   0          2d
[root@master ~]#
```

3. 如果Pod未正常运行，执行命令查看Pod启动失败原因,并进行修复

```
kubectl -n kube-plugin describe pod <podName>
```

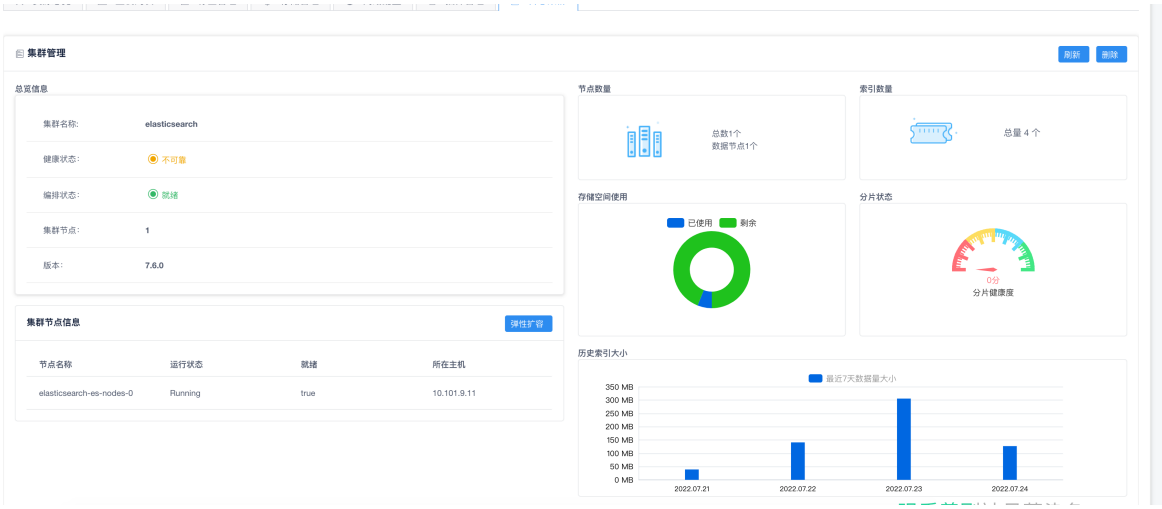
4. 查看prometheus容器的日志，检查是否正常运行

```
[root@master ~]# kubectl  -n kube-plugin logs -f prometheus-6544694566-9zj2w  prometheus
level=info ts=2022-07-22T09:39:20.796737435Z caller=main.go:225 msg="Starting Prometheus" version="(version=2.1.0, branch=HEAD, re
level=info ts=2022-07-22T09:39:20.79703715Z caller=main.go:226 build_context="(go=go1.9.2, user=root@6e784304d3ff, date=20180119-1
level=info ts=2022-07-22T09:39:20.797105237Z caller=main.go:227 host_details="(Linux 3.10.0-1160.el7.x86_64 #1 SMP Mon Oct 19 16:1
level=info ts=2022-07-22T09:39:20.797163584Z caller=main.go:228 fd_limits="(soft=1048576, hard=1048576)"
level=info ts=2022-07-22T09:39:20.80390663Z caller=main.go:499 msg="Starting TSDB ..."
level=info ts=2022-07-22T09:39:20.805869768Z caller=web.go:383 component=web msg="Start listening for connections" address=0.0.0.0
level=info ts=2022-07-22T09:39:20.816198539Z caller=main.go:509 msg="TSDB started"
level=info ts=2022-07-22T09:39:20.81623494Z caller=main.go:585 msg="Loading configuration file" filename=/etc/prometheus/prometheu
level=info ts=2022-07-22T09:39:20.818311902Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.819219489Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.819748737Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.820282634Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.820799091Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.821387867Z caller=kubernetes.go:191 component="discovery manager scrape" discovery=k8s msg="Usin
level=info ts=2022-07-22T09:39:20.822875315Z caller=main.go:486 msg="Server is ready to receive web requests."
level=info ts=2022-07-22T09:39:20.823248559Z caller=manager.go:59 component="scrape manager" msg="Starting scrape manager..."
```

2.3 集群日志集群信息无法查看

1. 检查多云底座控制台，集群插件是否安装

2. 插件日志集群页面，检查日志集群是否正常



3. 如果集群状态不正常，登录集群master节点，执行命令查看ES集群状态

```
[root@master ~]# kubectl -n kube-plugin get es
NAME            HEALTH   NODES   VERSION   PHASE   AGE
elasticsearch   yellow   1       7.6.0     Ready   27h
[root@master ~]#
```

4. 如果集群PHASE状态不为Ready，执行命令查看详细原因

```
kubectl  -n kube-plugin describe es elasticsearch
```

5. 如果ES集群状态运行正常，可以使用ES接口查看ES详细状态

```
# ES集群默认账号密码 elastic:es_management

# 集群状态
[root@master ~]# curl -k -u "elastic:es_management"  "https://10.101.9.11:59200/_cat/health?v"
epoch      timestamp cluster       status node.total node.data shards pri relo init unassign pending_tasks max_task_wait_time acti
1658656650 09:57:30  elasticsearch yellow          1         1      4   4    0    0        4             0                  -
[root@master ~]#

# 索引状态
[root@master ~]# curl -k -u "elastic:es_management"  "https://10.101.9.11:59200/_cat/indices?v"
health status index               uuid                   pri rep docs.count docs.deleted store.size pri.store.size
yellow open   logstash-2022.07.24 YTuYZcuESc2JOR5RpMj_mA   1   1     311863            0    128.4mb        128.4mb
yellow open   logstash-2022.07.23 SFQDnyA-SXaIN-PM6gzFMQ   1   1     852210            0    306.1mb        306.1mb
yellow open   logstash-2022.07.22 DP2WxWyjTL2skWnaUL5R3w   1   1     399515            0    141.1mb        141.1mb
yellow open   logstash-2022.07.21 5mvHAB2UQ_yPajCqYebnYA   1   1     105310            0     39.8mb         39.8mb
[root@master ~]#

# 分片状态
[root@master ~]# curl -k -u "elastic:es_management"  "https://10.101.9.11:59200/_cat/shards?v"
index               shard prirep state        docs   store ip           node
logstash-2022.07.23 0     p      STARTED    852210 306.1mb 10.1.166.133 elasticsearch-es-nodes-0
logstash-2022.07.23 0     r      UNASSIGNED
logstash-2022.07.22 0     p      STARTED    399515 141.1mb 10.1.166.133 elasticsearch-es-nodes-0
logstash-2022.07.22 0     r      UNASSIGNED
logstash-2022.07.24 0     p      STARTED    311863 128.5mb 10.1.166.133 elasticsearch-es-nodes-0
logstash-2022.07.24 0     r      UNASSIGNED
logstash-2022.07.21 0     p      STARTED    105310  39.8mb 10.1.166.133 elasticsearch-es-nodes-0
logstash-2022.07.21 0     r      UNASSIGNED
[root@master ~]#

# 节点状态
[root@master ~]# curl -k -u "elastic:es_management"  "https://10.101.9.11:59200/_cat/nodes?v"
ip           heap.percent ram.percent cpu load_1m load_5m load_15m node.role master name
10.1.166.133           46          67   5    0.15    0.18     0.34 dilm      *      elasticsearch-es-nodes-0
[root@master ~]#
```

3. 应用管理异常处理

3.1 商店应用创建失败

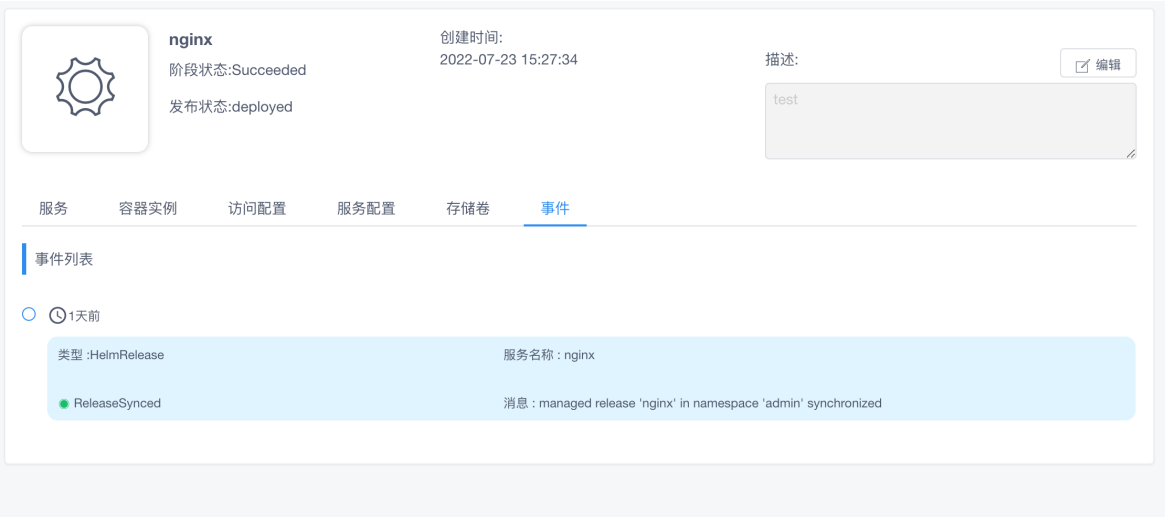
1. 登录部署应用所在的集群master节点，查看helm-operator的Pod是否运行正常

```
[root@dce dce]# docker ps |grep ftp
96e54dfb9418   bogem/ftp           "/usr/sbin/run-vsftp…"   4 days ago   Up 4 days       0.0.0.0:47400-47470->47400-47470/tcp, :::
[root@dce dce]#
```

2. 登录dce服务器查看dce服务日志

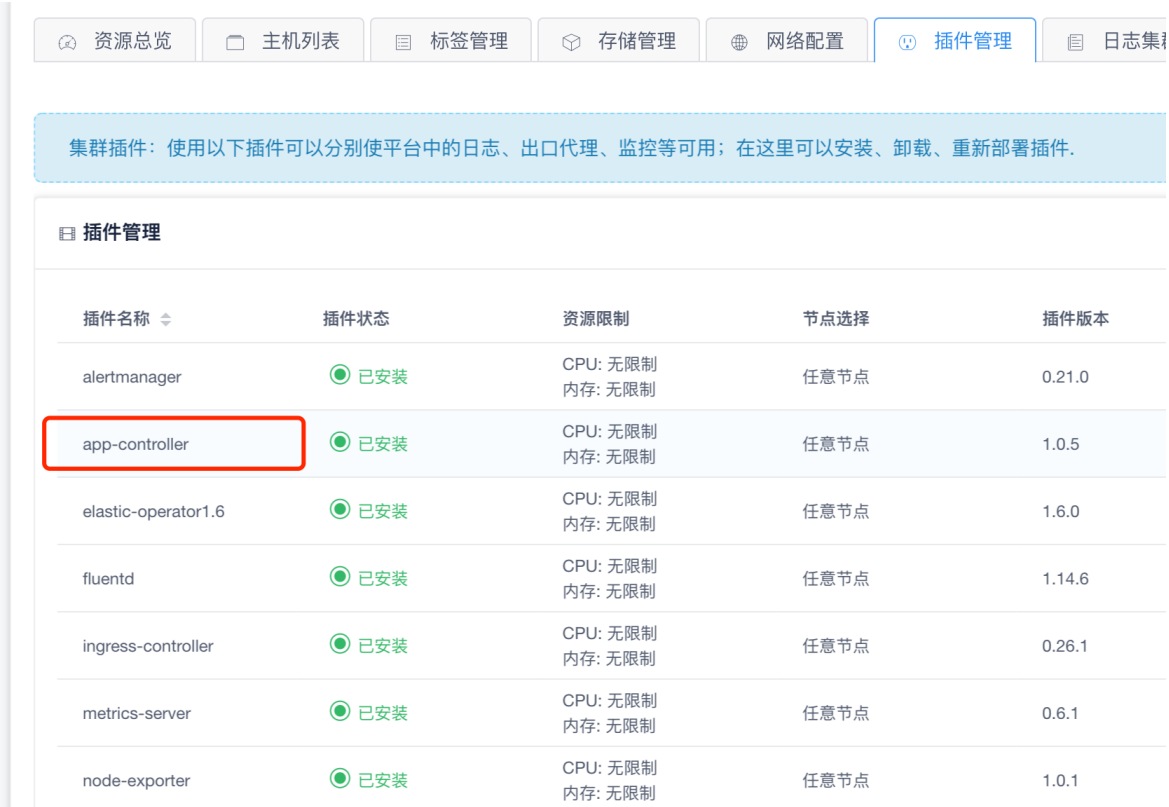
```
tailf /paas/dce/logs/api-server.log
```

3. 创建完成后，可以登录多云底座页面，点击商店应用名称，进入详情页，点击事件，查看应用部署过程记录



3.2 应用部署失败

1. 登录多云底座控制台，点击部署应用所在的集群详情页
2. 检查集群插件 app-controller是否正常安装



3. 登录集群master节点，查看app-controller的Pod是否正常运行

```
[root@master ~]# kubectl -n kube-plugin get pod |grep app
application-controller-manager-546f5b97b-w4m5n   2/2     Running   0          2d2h
[root@master ~]#
```

4. 查看application-controller的控制器日志，查看应用创建过程日志,查看是否存在报错信息

```
kubectl -n kube-plugin logs -f application-controller-manager-546f5b97b-w4m5n manager
2022-07-24T11:10:08.257Z        INFO    controller      found deployment has changed and updating by spec module.{"namespace": "ad
2022-07-24T11:10:08.257Z        INFO    controllers.Application reconcile svc...        {"application": "admin/app-a3ea90b1f05c",
2022-07-24T11:10:08.257Z        DEBUG   controller-runtime.manager.events       Normal  {"object": {"kind":"Application","namespac
2022-07-24T11:10:08.263Z        INFO    controllers.Application reconcile proxy...      {"application": "admin/app-a3ea90b1f05c",
2022-07-24T11:10:08.263Z        DEBUG   controller-runtime.manager.events       Normal  {"object": {"kind":"Application","namespac
2022-07-24T11:10:08.263Z        INFO    controllers.Application reconcile all done.     {"application": "admin/app-a3ea90b1f05c",
2022-07-24T11:10:08.263Z        DEBUG   controller-runtime.manager.events       Normal  {"object": {"kind":"Application","namespac
```